Password-Based Door Lock System

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Abstract

Security systems play a crucial role in modern infrastructure. Traditional lock-and-key mechanisms are vulnerable to duplication, theft, and unauthorized access. This paper presents the design and implementation of a password-based door lock system using Arduino. The system authenticates users through a 4×4 matrix keypad and grants access only when the correct password is entered. A servo motor simulates the door locking mechanism, while LEDs and a buzzer provide visual and audio feedback. The system includes a security feature that temporarily locks after three incorrect attempts. The project demonstrates a low-cost and efficient embedded security solution aligned with Sustainable Development Goal 9, promoting innovation and smart infrastructure.

Introduction

Security has become one of the most important requirements in modern homes, offices, industries, and institutions. Traditional lock-and-key systems are gradually being replaced by electronic security systems due to their improved reliability and flexibility. This project presents the design and implementation of a Password Based Door Lock System using Arduino.

The system uses a 4x4 matrix keypad to allow users to enter a password. The entered password is verified by the microcontroller. If the password matches the stored password, a servo motor rotates to unlock the door and a green LED indicates successful access. If the password is incorrect, a red LED glows and a buzzer alerts the user. After three wrong attempts, the system temporarily locks to prevent unauthorized access.

This project supports SDG 9 – Industry, Innovation and Infrastructure, as it promotes smart automation, digital security, and technological innovation in infrastructure systems.

Objective of the Project

The main objectives of this project are:

- To design a secure and low-cost electronic door locking system.
- To implement password authentication using a keypad.
- To display system status on a 16x2 LCD.
- To automate door movement using a servo motor.
- To enhance security with a limited number of password attempts.
- To demonstrate the practical implementation using Arduino in TinkerCAD.

The system aims to provide a simple yet effective security solution that can be implemented in homes, laboratories, and small offices.

SDG Goal Connection

This project supports SDG 9 – Industry, Innovation and Infrastructure by:

- Promoting smart infrastructure systems.
- Encouraging innovation in electronic security.
- Demonstrating automation using microcontroller technology.

System Overview

The Password Based Door Lock System consists of the following main components:

1. **Arduino UNO** – The main controller that processes input and controls output devices.
2. **4x4 Matrix Keypad** – Used to enter the password.
3. **16x2 LCD Display** – Displays messages such as "Enter Password", "Access Granted", and "Access Denied".
4. **Servo Motor** – Simulates door lock mechanism.

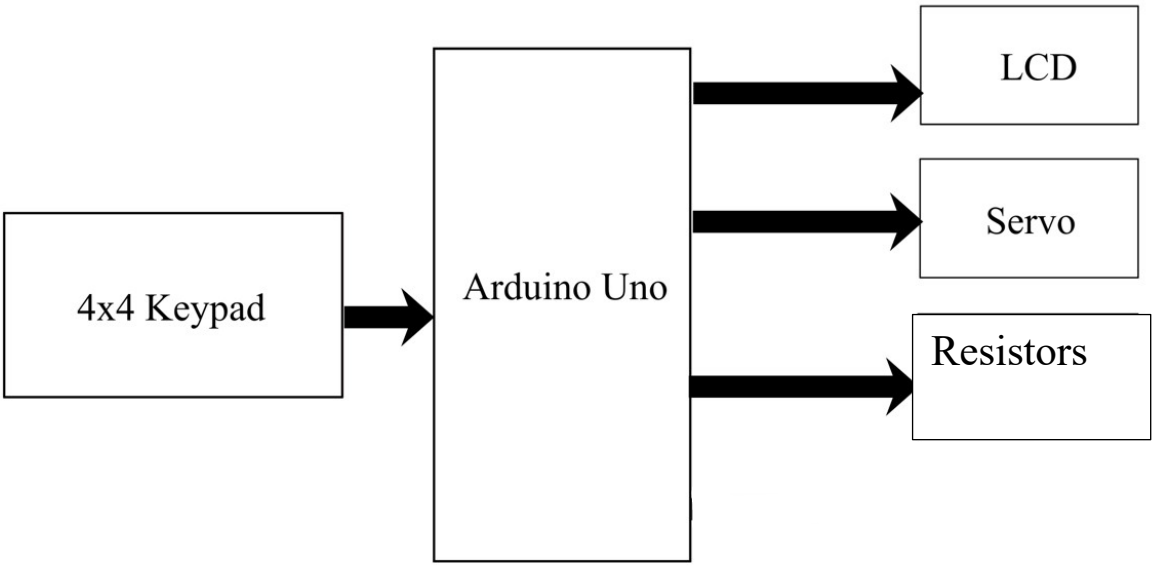

Fig: - Block Diagram

Working Principle:

1. The system prompts the user to enter a password.
2. The user enters a 4-digit password using the keypad.
3. The Arduino compares the entered password with the stored password.
4. If correct:
    - LCD displays “Access Granted”.
    - Servo motor rotates to 90° (unlock).
    - After 3 seconds, servo returns to 0° (lock).
5. If incorrect:
    - LCD displays “Access Denied”.
6. After 3 wrong attempts:
    - System displays “System Locked”.
    - Access is disabled for 5 seconds.

This creates a reliable electronic locking system.

Hardware Description

➢ **Arduino UNO**

The project is built around the Arduino UNO. It uses the ATmega328P microcontroller and provides digital and analog input/output pins. It processes keypad input and controls LEDs, buzzer, LCD, and servo motor.

➢ **16x2 LCD**

The 16x2 LCD displays two rows of 16 characters each. It provides real-time feedback to the user.

➢ **4x4 Keypad**

The keypad consists of 16 keys arranged in 4 rows and 4 columns. It works on matrix scanning principle.

➢ **Servo Motor**

The servo rotates between 0° and 90° to simulate door locking and unlocking.

Software Design

The system is programmed using Arduino IDE (simulated in TinkerCAD). The following libraries are used:

- LiquidCrystal.h – Controls LCD
- Keypad.h – Handles keypad input
- Servo.h – Controls servo motor

The program logic:

- Initialize LCD and servo
- Continuously check keypad input
- Store entered digits
- Compare with stored password
- Execute actions accordingly

The system uses conditional statements (if-else) and loops to manage operations efficiently.

Advantages

- Low cost
- Easy to implement
- User-friendly interface
- Improved security
- Prevents unauthorized access
- Expandable to IoT systems

Applications

- Home security systems
- Office access control
- Bank lockers
- Laboratories
- Smart buildings
- Industrial equipment protection

Limitations

- Fixed password (unless change feature is added)
- Not encrypted (basic implementation)
- No remote monitoring
- Limited to 4-digit password

Future Enhancements

The system can be improved by:

- Adding RFID authentication
- Integrating fingerprint sensor
- Connecting to IoT (Blynk / WiFi module)
- Adding mobile app control
- Implementing OTP verification
- Using EEPROM for password storage

Conclusion

The Password Based Door Lock System is a simple and effective security solution implemented using Arduino. It demonstrates the integration of hardware components like keypad, LCD, servo motor, LEDs, and buzzer to create an automated locking system.

The project enhances understanding of embedded systems and real-time control. It aligns with modern smart infrastructure development and supports SDG 9 by promoting technological innovation in security systems.

This project can serve as a foundation for advanced security applications in homes and industries.